Shivang Tripathi

Ph.D. · Postdoc (USA) · B.Tech (ECE)

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Citations: 1645 | **h-index:** 7 | **i10-index:** 5 | *(Google Scholar, May 2026)*

Interdisciplinary researcher and faculty with **10+ years** of experience spanning **experimental physics, electronics engineering, and computing sciences**. Research background covers high-energy physics detector instrumentation (Belle II, EIC), FPGA firmware design, SiC radiation sensors, and Monte-Carlo simulation. Currently faculty in the **Dept. of AI & Data Engineering (AIDE), IIIT Kota**, teaching and mentoring across **Quantum Computing, Cryptography & Cybersecurity, Applied AI, and Mathematical/Statistical Foundations for Data Science**, bridging deep hardware expertise with modern CS and AI frontiers.
- Postdoctoral Fellow, University of Hawaiʻi at Mānoa (2020–2024).
- Open to permanent faculty positions in ECE, Physics, CSE, or AIDE departments.

Professional Experience

Guest Faculty, Dept. of AIDE, *IIIT Kota* — Aug 2025 – Present
Rajasthan, India

- Teaching UG/PG courses: Cryptography & Cybersecurity, Applications of AI, Statistics for Data Science (+ Lab), Quantum Computing, Mathematical Foundations for Data Science (M.Tech & PhD)
- Serving as Faculty Coordinator, **Training & Placement Cell**
- Mentoring UG final-year projects in AI, ML, DL, FPGA design, and embedded systems

Faculty, Dept. of ECE, *IIIT Ranchi* — Aug 2024 – Jun 2025
Jharkhand, India

- Taught Python Programming (3 sections, UG), Digital Logic Design, Advanced Embedded Processor & Microcontrollers, VLSI-MEMS Lab
- ERP Coordinator for Hostel module implementation (Samarth platform)
- Faculty Advisor for CSE 4th-year batch

Postdoctoral Research Fellow, *University of Hawaiʻi at Mānoa, IDLab* — Jan 2020 – May 2024
Honolulu, HI, USA

- **Belle II / SuperKEK (Japan):** Contributed to Belle II iTOP Cherenkov detector daily operations, system maintenance, and hardware upgrades; Co-authored the iTOP performance and calibration paper (*NIM-A*, 2025)
- **EIC-PID Readout Electronics:** Designed and fabricated SiREAD-ASIC-based Daughter Cards (DCs) capable of processing 64 channels of 256-anode MAPMTs/SiPMs; developed Spartan-6 FPGA firmware using Quad-Byte Link and Gigabit Transceiver protocols
- **HMB Calorimeter Readout:** Designed an integrated firmware/software framework for a NaI(Tl)-based calorimeter readout system, utilizing Xilinx Vivado/Vitis to deploy designs onto a Zynq SoC-FPGA (Avnet MicroZed) platform.
- Was Associated with the eRD-14 EIC-PID consortium (BNL) and Belle II TOP detector group

Education

Ph.D., Engineering Sciences, *Homi Bhabha National Institute (HBNI)* — 2013 – 2020
Mumbai, India

- *Thesis:* Study of SiC-Based Neutron Detector for Applications in the Harsh Environment of Fast Reactors
- *Supervisor:* Prof. Dr. K. Devan, Head RND, IGCAR, Kalpakkam

B.Tech.(*Hons.*), Electronics & Communication Engineering, *Lovely Professional University* — 2007 – 2011
Punjab, India

- *CGPA: 8.32 / 10*

Research Highlights

HEP Instrumentation	Design of integrated readout electronics for Belle II (iTOP) and the Electron-Ion Collider (RICH/PID detectors); firmware & software for DAQ systems utilizing Xilinx SP6, Zynq SoC, and PYNQ FPGA platforms
Radiation Detectors	GEANT4 Monte-Carlo modeling of planar & 3D-stacked SiC neutron detectors; converter material optimization (HDPE, LiH, Perylene, PTCDA) for Fast Reactors
Semiconductor Simulation	TCAD modeling of 4H-SiC Schottky Barrier Diodes; irradiation-effects study (neutron, proton, γ, electron) on electrical characteristics

Publications

Peer-Reviewed Journal Articles

[J1] Shivang Tripathi *et al.*, "Performance simulation of the LiH-SiC-based Fast Neutron Detector using Geant4 and TCAD," *NIM-A* **916**, 246–256 (2019). DOI [Q2]

[J2] Shivang Tripathi *et al.*, "Effect of electron and proton irradiation on SiC-based fast neutron detectors," *JINST* **14**, P02002 (2019). DOI [Q2]

[J3] Shivang Tripathi *et al.*, "Towards radiation hard converter materials for SiC-based Fast Neutron Detectors," *JINST* **13**, P05026 (2018). DOI [Q2]

[J4] Shivang Tripathi *et al.*, "Enhancement in planar fast neutron detector efficiency with stacked structure using Geant4," *Nucl. Sci. Tech.* **28**, 154 (2017). DOI [**Q1**, IF: 3.6]

Major Collaboration Papers (as contributing author)

[C1] Belle II Collaboration, *et al.*, "The Imaging Time-of-Propagation Detector at Belle II," *Nucl. Instrum. Methods Phys. Res. A* **1080**, 170627 (2025). DOI [Q2]

[C2] EIC Collaboration, "Science Requirements and Detector Concepts for the EIC: EIC Yellow Report," *Nucl. Phys. A* **1026**, 122447 (2022). DOI [Q2]

[C3] Belle II Collaboration, "Belle II Executive Summary," arXiv:2203.10203 (2022).

[C4] EIC Collaboration, "CORE – a COmpact detectoR for the EIC," arXiv:2209.00496 (2022).

Book Chapter

[B1] Shivang Tripathi *et al.*, "Investigation of Perylene as a Converter Material for Fast Neutron Detection Using GEANT4," in *Advances in Systems, Control and Automation*, LNEE vol. 442, Springer Singapore (2018). DOI

Conference Presentations (Selected)

- Effect of gamma irradiation on SiC-based FNDs. *RSM-MSENM-2018, IGCAR, India* (Oral)
- SiC neutron detector for power-range neutron flux monitoring. *IAEA Technical Meeting, Vienna, Austria* (Oral, 2017)
- LiH-SiC based Fast Neutron Detector for harsh environment. *IARPIC-2018, BARC, Mumbai* (Oral)
- TCAD-assisted analysis of SiC-based Fast Neutron Detector. *12th IEEE NMDC-2017, Singapore* (Oral)
- Geant4 simulations of SiC detectors for fast neutron spectroscopy. *IEEE INDICON-2015, India* (Oral). DOI

Technical Skills

HDL / Firmware	VHDL, Verilog; Xilinx Vivado, Vitis, ISE, ChipScope; Zynq SoC, PYNQ, Spartan-6
Programming	Python, C, C++, MATLAB, TCL-tk, ROOT
EDA & PCB	Altium Designer, Cadence Virtuoso, Modelsim
Semiconductor TCAD	SILVACO ATLAS, DECKBUILD, DEVEDIT, Victory 3D
Monte-Carlo	Geant4, SRIM/TRIM, DD4Hep, ROOT
OS / Other	Sci-Linux, Origin Pro, LaTeX

Awards, Scholarships & Fellowships

2020–2023	Instrumentation Innovation Research Fellowship, University of Hawaiʻi
2013–2019	Research Fellowship, Dept. of Atomic Energy (DAE), Govt. of India
2014, 2016	Qualified GATE (Graduate Aptitude Test in Engineering)

Academic & Professional Service

2020–2024	Member, TOP Detector Group, Belle II Experiment, SuperKEK, Japan
2020–2022	Member, eRD-14 EIC-PID Consortium, Brookhaven National Laboratory, USA
Active	IEEE Member
2021	Online Certification in Data Analysis using Python, University of Pennsylvania
2022	Conducted GEANT4 simulation workshop for BARC trainees
Jul–Dec 2011	Internship, DKOP Labs Pvt. Ltd. – Design & simulation of 10 Gb Ethernet transmitter in Verilog

UG Project Mentoring (2025)

Devanshu Gaur	Multi-protocol serial communication core using HDL
Rohan Tiwari	Custom HW design on Xilinx PYNQ for image processing with AXI integration
Samarth Joshi	PCB defect detection using YOLO and Deep Learning
Saxena, Tripathi, Kumar	Python-based AI game development
Ashwin Kumawat	Tigsaw: AI-powered web widgets for smart optimization

References

Prof. Dr. K. Devan (PhD Supervisor)
Professor & Head RND, IGCAR / HBNI, Kalpakkam
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Shri C. P. Nagaraj
Former Head, Nuclear Instrumentation Section, IGCAR
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Dr. Shashikant Sharma
Assistant Professor, IIIT Ranchi
sksharma@iiitranchi.ac.in

Prof. Tom Browder (Postdoc Supervisor)
Dept. of Physics & Astronomy, UH Mānoa
teb@phys.hawaii.edu

Dr. Mudit Mishra (*Now with SLAC*)
Research Staff, IDLab, UH Mānoa
mmishra9@hawaii.edu

Late Prof. Gary Varner (*Postdoc Supervisor till July 23, 2023*)
IDLab, UH Mānoa

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